Task 3: Jurisdiction-Aware AML Alert Calibration Tool

For Task 3, I designed a jurisdiction-aware AML alert calibration tool for Citibank, N.A. / Citi Singapore. The aim of the tool is to show how the same set of transactions may produce different compliance alerts depending on whether the active regulatory setting is the United States or Singapore.

I used synthetic transaction data because real bank transaction data is not available for this assignment. The dataset contains 1,000 simulated transactions. Each transaction includes fields such as transaction amount, currency, origin and destination country, customer risk level, sanctions match score, whether the counterparty is new, recent transaction frequency, whether beneficial ownership information is complete, whether wire transfer information is complete, and whether the transaction has a known suspicious pattern.

The tool applies two different alert configurations. In US mode, the scoring gives more weight to sanctions match risk, USD transactions, US nexus, and structuring-like patterns. This reflects the US focus on suspicious activity reporting, BSA/AML monitoring, and OFAC sanctions risk. In Singapore mode, the scoring gives more weight to customer due diligence issues, beneficial ownership completeness, wire transfer information, high-risk customers, and STRO-related review indicators. This reflects the MAS AML/CFT framework and Singapore's suspicious transaction reporting structure.

The main metric I monitor is alert rate. I also look at false-positive proxy rate, known suspicious pattern capture, and estimated manual review workload. In the baseline analysis, US mode generated 79 alerts out of 1,000 transactions, while Singapore mode generated 107 alerts. US mode captured 26 out of 49 known suspicious patterns, and Singapore mode captured 27 out of 49. This suggests that the Singapore configuration was slightly more sensitive in this dataset, but also created more manual review work.

I also tested how the results changed when assumptions shifted. Lowering the US sanctions threshold from 0.75 to 0.65 increased US alerts from 79 to 88, but did not capture more known suspicious patterns. This suggests that lowering the threshold mainly added workload rather than improving detection. When high-risk-country exposure was increased, both US and Singapore alerts increased, with Singapore alerts rising slightly more.

The tool does not automatically file SARs or STRs, decide whether to offboard customers, or give legal advice. I made this boundary choice because suspicious transaction reporting still requires human judgement. The tool is meant to support compliance teams by making jurisdiction-specific trade-offs clearer, not to replace their decisions.